Shell - MultiAgent Financial Sysytem

IFT0001 - C

Introduction and Objective

The rapid advancement of Artificial Intelligence has fundamentally transformed the financial sector, automating complex tasks that once required substantial manual effort. This project developed a unified AI agent system designed to assist the workflow of an asset manager. By integrating specialized sub-agents into a parallel architecture, this system addresses the chronic challenge of aggregating disparate data from news, financial statements, and price histories.

The architecture is divided into two primary analytical agents: fundamental and technical. Fundamental agents evaluate financial health through ratio analysis, competitor benchmarking, and fair value calculations to determine intrinsic value. Conversely, technical agents assess market psychology and timing using indicators like RSI, MACD, and ADX to provide a directional bias.

The Aggregator Agent serves as the system's core. It mimics a human investment committee by synthesizing conflicting data into a nuanced, standardized investment memo. This process replaces labor-intensive manual research with automated data ingestion pipelines, allowing for the instant retrieval of financial statements and historical price action.

Ultimately, this multi-agent structure functions as a scalable decision-support tool for portfolio managers. By anchoring qualitative debates in rigorous quantitative valuation and incorporating a backtesting engine, the system ensures that every asset is evaluated against consistent, objective criteria. This reduces analyst bias and significantly lowers the cognitive load required to combine qualitative insights with quantitative signals, resulting in a more efficient and comprehensive investment strategy.

Data Collection and Preparation

Data Sources

The agent fetches data from three primary sources to ensure a comprehensive analysis that covers both historical financial performance and current market momentum:

- For fundamental data and dividend information, the **Alpha Vantage API** is leveraged to collect financial statements spanning the past five years of the target companies.
- Technical data and market information are retrieved via the **yfinance library**. The system fetches historical OHLCV data for Shell and its primary competitors to calculate key technical indicators.
- To improve accuracy of the DCF model, the Fama-French three factors are retrieved from the **Kenneth R. French Data Library** using pandas_datareader.

Cleaning and Validation

The agent applies strict preprocessing steps after retrieving the raw data to maintain data completeness and consistency. First, converting strings to numbers is broadly applied to ensure all financial data is compatible with numerical operations, as API responses often return numerical values as strings. If any non-numerical content occurs, it is automatically converted to "NaN" to prevent runtime errors.

Secondly, it standardizes null values by mapping inconsistent data (e.g., "NA", "None", or spaces) to a standard format, while dropping non-available values is employed to remove invalid columns. A final layer of filling of non-available values with "N/A" is applied to ensure the AI correctly identifies missing values rather than misinterpreting them as calculation errors.

Next, temporal data alignment is essential as fiscal period end dates do not usually coincide with active trading days. The agent leverages the merging function with backward filling to ensure the system fetches the most recent stock price on or prior to the fiscal period end date, which avoids look-ahead bias. Lastly, it computes the financial ratios and technical indicators and rounds them to four decimals as input to avoid AI agent confusion.

Figure 1: Fundamental Schema

comp	fiscalDateEnding	GrossMargin	OperatingMargin	NetProfitMargin	ROE
SHEL	2020-12-31	-0.0720	-0.1414	-0.1201	-0.1396
SHEL	2021-12-31	0.1371	0.0852	0.0769	0.1169

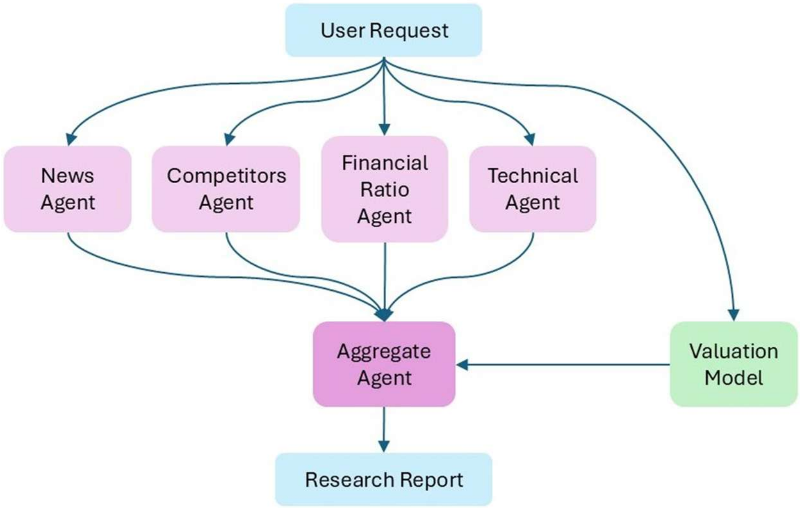
Figure 2: Technical Schema

	Close	Volume	RSI	MACD_Line	MACD_Signal
Date					
2025-10-06	73.5696	4018859	61.3016	0.2648	0.0837
2025-10-07	74.2035	4000764	63.9281	0.4305	0.1530

Agent Design and Architecture

This AI agent system is designed as an LLM based multi-agent architecture, as illustrated in Figure 3. By structuring the operational workflow, the system isolates each agent's focus to optimize analysis results. The following sections explain the details of each component:

Figure 3: AI Agent Structure



LLM Layer

Gemini models act as reasoning and reporting engines rather than numerical calculators. The models utilized include Gemini 2.5 Flash and Gemini 2.5 Flash Lite. Through these models, several specialist agents were established to generate focused domain insights, which include news, competitors, financial ratios, and technical analysis. Finally, an aggregator agent synthesises all outputs into a single investment memo.

Tools

Several tools were used for each specific agent. By processing and selecting the data needed for each agent, these tools ensure the agents acquire correct and effective data inputs. Apart from the custom tools, the news agent features a built-in tool "google_search", which provides information from the Google search engine to include up-to-date data. Agents validate tool outputs and halt execution on errors to ensure robustness.

Prompt Design and Agent Logic

Each agent is governed by a role-specific prompt to avoid confusion and inconsistency in the AI model. These prompts clearly define the scope, state the analytical focus, and provide the output structure. Using each sub-agent's analysis, the aggregator agent explicitly addresses conflicting signals across fundamentals, technicals, news, and valuation.

Execution Flow and Orchestration

Initially, user input triggers data ingestion and feature engineering in Python. Specialist agents then run concurrently using a parallel structure, isolating responsibilities and reducing latency. Apart from the sub-agents, the DCF model executes independently as it does not require further AI analysis. Finally, an aggregator agent integrates all evidence, including the DCF valuation, into a concise portfolio-manager style investment memo.

Methodology and Implementation

News Agent

The News Agent monitors and analyzes the latest news related to Shell. Recognizing that most information is rapidly reflected in stock prices, the data retrieval window is limited to two weeks. By utilizing the Google Search tool, the agent first establishes a foundational understanding of the company before specifically searching for news events that could impact stock price. Subsequently, it generates a descriptive memo based on the aggregated information.

Competitors Agent

To facilitate a direct comparison with Shell's competitors, this agent identifies three main companies within the same industry. Using the financial ratios retrieved and calculated via Alpha Vantage, the latest data for both Shell and its competitors is provided as input for the agent to analyze and formulate comparative conclusions.

Financial Ratio Agent

Similar to the Competitors Agent, the Financial Ratio Agent retrieves the financial ratios via Alpha Vantage. The key distinction is that it focuses exclusively on Shell, spanning the past five years. Utilizing this data, the agent analyzes the trends across four main pillars: operational efficiency, profitability and margins, financial solvency and health, and growth and valuation. Furthermore, it highlights inconsistencies if divergences among the metrics are detected. Finally, the agent generates a structured report.

Technical Agent

For the Technical Agent, the system ingests historical OHLCV data from Yahoo Finance using the yfinance library. To compute technical indicators, the pandas_ta library is utilized as the primary analytical package. By incorporating multiple indicators, the agent analyzes the confluence and shifts across trading sessions, focusing on the trend, momentum, and volatility. At last, it synthesizes these findings into a cohesive technical memo.

Valuation Model

For the stock price valuation, this report utilizes a Discounted Cash Flow (DCF) model. The process began with calculating the Free Cash Flow (FCF). We then determined the average FCF growth rate over the historical period using a weighted average approach, assigning greater weight to more recent cash flows to reflect current operational trends. The second phase involved calculating Weighted Average Cost of Capital (WACC), where the cost of equity was derived via the Fama-French three-factor model and cost of debt was sourced from current financial data. In the third phase, we established a terminal growth rate of 2%, assuming that starting from the sixth year, Shell's FCF will grow at this constant rate in perpetuity. In the final step, we applied these inputs to derive Shell's enterprise value, which was subsequently converted to an estimated share price.

Aggregate Agent

The Aggregate Agent synthesizes the analyses from four specialist agents alongside the DCF valuation results into a cohesive 300-word investment report. Adhering to a predefined structure, the agent generates a memo that includes a target price, executive recommendations, and a balanced assessment of advantages and disadvantages. Furthermore, it weighs the relative importance of the information and reconciles conflicting signals across the inputs to produce an objective and decisive recommendation.

Backtest

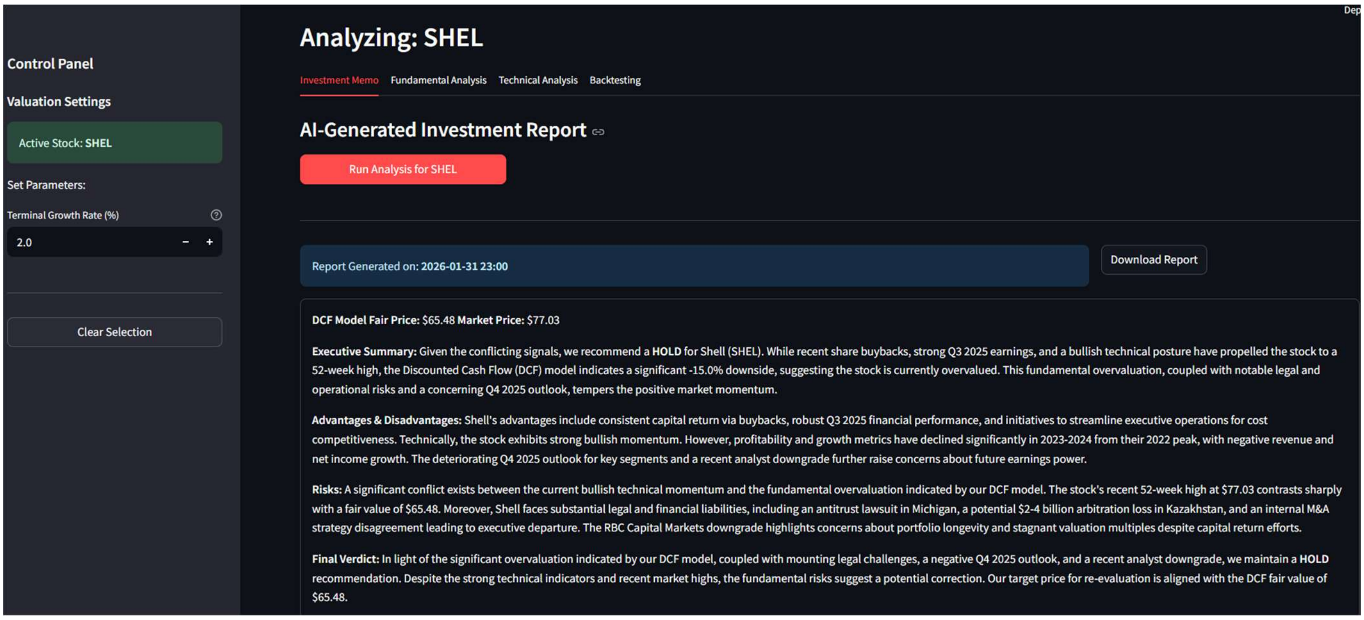
In addition to the current analysis, we conducted backtesting on the model to evaluate its historical performance. For instance, on December 31, 2024, the AI agent assigned a BUY rating to Shell (SHEL) when the share price was trading at approximately \$59. Over the subsequent year, the stock appreciated to exceed \$70, demonstrating the accuracy of the agent's investment advice. Similarly, on December 31, 2023, the agent issued a HOLD rating when the stock was trading at approximately \$61. A year later, the share price had consolidated around the \$60 level, further confirming the precision of the agent's recommendations.

Result and Analysis

Summary of Agent Outputs

Figure 4 illustrates the AI agent's user interface and the comprehensive investment memo it generates. The system is designed with several core functionalities. Initially, users manually input the stock ticker of the target company and initiate the process by clicking "run analysis". The agent then automatically generates a comprehensive investment memo synthesized from technical indicators, fundamental metrics, and DCF valuation model. This memo includes a consolidated analysis of the financial data, an estimated share price derived from the valuation model, and a definitive investment rating (Buy, Hold, or Sell).

Figure 4: Investment Memo



In addition to the primary memo, users can access details by navigating the tabs beside the investment memo (as shown in Figure 5). This section presents supplementary data and functionalities, including corporate fundamental metrics, technical analysis charts, and backtesting capabilities. The backtesting feature is particularly notable, as it allows users to simulate specific past dates to evaluate how the agent's functions and predictive accuracy would have performed historically.

Figure 5: Analysis Chart

Interpretation of Results

Based on the data available at the time of this report, the AI agent assigned a **HOLD** rating to Shell (SHEL). The agent identified favorable signals in the technical analysis, specifically highlighting an optimistic long-term trend in the share price alongside positive short-term momentum.

However, the agent flagged several warning signs within the fundamental analysis. While it acknowledged positive factors, such as Shell's consistent execution of share repurchases, robust liquidity, and improving inventory management, these were counterbalanced by significant concerns. The agent noted that Shell's profitability and growth metrics exhibited a marked decline in 2023 and 2024. Furthermore, the company faces potential headwinds from a possible liability of $2–4 billion related to an international arbitration case involving the Kazakhstan field, as well as a recent downgrade by RBC Capital Markets, both of which could negatively impact future share performance.

Finally, relying on the results of our Discounted Cash Flow (DCF) model, the agent identified that the current market price of $77.03 is higher than the model's intrinsic valuation of $65.48, suggesting that the stock may be overvalued at present. Synthesizing these factors, the agent concluded with a comprehensive **HOLD** recommendation.

Discussion and Limitations

Through utilizing the AI agent, we could easily produce an investment report within a minute. While it appears to be a convenient tool for improving efficiency and simplifying the process of financial analysis, there are some constraints that are essential to consider when applying it:

AI Judgement Reliability

Currently, AI models are competing intensively and are becoming increasingly powerful. However, despite rapid development, AI can still generate questionable or even false information. This makes AI-generated content less reliable, especially in a relatively subjective field like finance. It is difficult to mitigate this issue unless AI evolves to a significantly higher level of sophistication. Therefore, we must incorporate human judgement into any AI-generated report to ensure accuracy.

Data Integrity

With restricted resources, we only have limited access to a few data sources. The data fetched from these sources might be incomplete or contain errors. As data is the core input for the AI agent to make judgements, data integrity becomes critical. To ensure data integrity, it is crucial to partner with a reliable data provider or even establish a proprietary database.

Token and Request Limitations

Similar to the constraints on data sources, there are rate limits on AI models due to a constrained budget. For our AI agent, the main issue arises from the request limit. By diversifying the models under the Google AI framework, we expand availability. Nevertheless, the system can currently only run three times per day.

Conclusion and Reflection

Key Insights

The system demonstrates that AI functions most effectively as a reasoning and reporting engine rather than merely a computational tool. By leveraging Python for deterministic data processing alongside Large Language Models (LLMs) for contextual analysis, asset managers can achieve superior transparency and consistency in decision-making processes.

The deployment of specialized agents facilitates a holistic assessment of an asset, enabling the Aggregator Agent to systematically reconcile conflicting signals, such as the divergence between robust technical indicators and weakening fundamentals. Furthermore, the workflow strategically segregates risk diagnosis from the core investment recommendation. This structural separation is crucial for institutional-grade asset management, ensuring objective and unclouded portfolio oversight.

Finally, by integrating "Back-to-Date" functionality, the system empowers managers to validate AI logic against historical outcomes. This capability is essential for fostering the algorithmic trust required to adopt AI solutions within high-stakes financial environments.

Possible Extension and Improvement

Data Integrity and Proprietary Integration: The current system relies on public data sources, which may be prone to incompleteness or inconsistencies. To mitigate this issue, future developments could involve integrating premium data providers, such as Bloomberg, or establishing a proprietary database. This would ensure higher data accuracy and enable the incorporation of alternative data streams for more granular analysis.

Dynamic Valuation and Stress Testing: To enhance the robustness of the system, the DCF model could be expanded to include Monte Carlo simulations and dynamic scenario analysis. For instance, quantifying the valuation impact of a 1% increase in interest would provide a probabilistic range of outcomes rather than a static point estimate, offering a more nuanced view.

Scalability and Latency Optimization: To address API rate constraints and optimize operational cost, the system architecture could be refined to batch independent requests or deploy private LLM instances. These improvements would facilitate high-frequency analysis across larger investment portfolios while significantly reducing latency.

Appendix II: Financial Ratios

Category	Financial Ratio	Calculation
Profitability	Gross Margin	Gross Profit / Total Revenue
	Operating Margin	Operating Income / Total Revenue
	Net Profit Margin	Net Income / Total Revenue
	Return on Equity (ROE)	Net Income / Total Equity
	Return on Assets (ROA)	Net Income / Total Assets
Leverage	Debt-to-Equity	Total Liabilities / Total Equity
	Current Ratio	Current Assets / Current Liabilities
	Interest Coverage Ratio	EBIT / Interest Expense
	Debt-to-EBITDA	Total Liabilities / EBITDA
Efficiency	Asset Turnover	Total Revenue / Total Assets
	Inventory Turnover	Cost of Goods Sold / Inventory
	Account Receivable Turnover	Total Revenue / Account Receivables
	Account Payable Turnover	Cost of Goods Sold / Account Payable
Growth	Revenue Growth	(Current Total Revenue / Previous Total Revenue) - 1
	Net Income Growth	(Current Net Income / Previous Net Income) - 1
	EPS Growth	(Current EPS / Previous EPS) - 1

Category	Financial Ratio	Calculation
Growth	FCF Growth	(Current Free Cash Flow / Previous Free Cash Flow) - 1
Valuation	Dividend Yield	Dividends / Stock Close Price
	P/E Ratio	Stock Close Price / EPS
	P/S Ratio	Stock Close Price / Revenue per Share
	P/B Ratio	Stock Close Price / Equity per Share

Appendix III: Technical Indicators

Technical Indicators	Explanation
Close Price	Stock’s close price adjusted for corporate actions.
Volume	Daily trading volume in the market.
Relative Strength Index (RSI)	Tell investors if the price is at an extreme point.
Moving Average Convergence Divergence (MACD)	Help traders find the start of a new trend.
Average Directional Index (ADX)	Measure how strong a market trend is.
Money Flow Index (MFI)	Similar to the RSI, but it includes trading volume.
Moving Averages (MA)	Average of the stock’s close price over 20, 50, and 200 days.
Support and Resistance (S&R)	Help traders spot potential reversals or breakouts for entry/exit points.
Volatility	Standard deviation of the stock’s daily return.